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Deep Convolutional Neural Network with Transfer Learning for Digital Art Style Classification

Computer Vision Project Report

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Abstract

Digital art is becoming more common on online platforms, and many artworks belong to different artistic styles. Identifying these styles manually can take time, especially when paintings from different styles have similar colours, textures, and patterns. This project focuses on using deep learning to classify digital artworks according to their artistic styles.

The project uses the WikiArt-10K dataset, which contains artworks associated with different styles. The proposed approach explores Convolutional Neural Networks (CNNs), residual connections, attention mechanisms, and transfer learning. These techniques can help a model learn useful features from artwork images.

The main aim of this project is to explore how deep learning can be used for art-style classification and to understand the challenges involved in building and evaluating such a system.

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Introduction

Background

Nowadays, many paintings and artworks are available on digital platforms. These artworks belong to different styles, such as Cubism, Impressionism, Pop Art, and Surrealism. Classifying them according to their style can help organise large art collections and make it easier for people to find artworks they are interested in.

However, identifying an artwork's style is not always easy. Two paintings from different styles may have similar colours or textures, while paintings from the same style may look very different. This makes art-style classification an interesting problem in computer vision.

Convolutional Neural Networks (CNNs) are commonly used for image classification because they can learn patterns directly from images. In this project, we explore CNNs along with residual connections, attention mechanisms, and transfer learning to understand how these techniques can help with art-style classification.

Problem Statement

The main problem addressed in this project is how to classify digital artworks into their correct artistic styles. Different styles can share similar visual features, which can make it difficult for a model to distinguish between them correctly.

Some of the main challenges are:

- Different styles may use similar colours and textures.
- Paintings belonging to the same style can look quite different.
- A model needs correctly labelled images to learn useful patterns.
- A model may perform well on training images but struggle with images it has not seen before.

This project explores a deep learning approach to study these challenges and understand how image classification can be evaluated.

Objectives

The main objectives of this project are:

1. To understand how CNNs can be used for digital art-style classification.
2. To study residual connections and attention mechanisms in image classification.
3. To understand the role of transfer learning.
4. To prepare and organise an image dataset for model evaluation.
5. To evaluate classification performance using suitable metrics.
6. To identify limitations and possible improvements.

Scope of the Project

This project focuses on classifying artworks according to their artistic styles. It covers dataset preparation, image preprocessing, the proposed deep learning approach, and model evaluation.

The project also discusses how attention mechanisms, residual learning, and transfer learning may help a model learn useful image features. The effectiveness of these techniques needs to be established through appropriate experiments.

Literature Review

Reference Research Paper

The main reference for this project is the research paper on digital art-style classification

The paper explains how deep learning techniques can be used to classify artworks based on their artistic styles. It provides the foundation for understanding the model architecture, dataset preparation, and evaluation process used in our project.

We selected this paper because identifying artistic styles can be challenging due to similarities in colours, textures, and visual patterns. By studying this research, we aim to understand how convolutional neural networks, residual connections, attention mechanisms, and transfer learning can help improve image classification.

Theoretical Background

Convolutional Layer

A convolutional layer applies small filters to an image or feature map. These filters learn patterns such as edges, shapes, and textures.

As the image passes through different layers, the network can combine simple patterns into more complex features. These learned features are then used by later layers for classification.

Activation Function

An activation function helps a neural network learn non-linear relationships. One commonly used activation function is the Rectified Linear Unit (ReLU).

It is defined as:

$$f(x) = \max(0, x) \tag{1}$$

ReLU keeps positive values and changes negative values to zero. It is commonly used in CNN architectures.

Pooling

Pooling reduces the spatial size of feature maps. This can reduce the amount of computation needed by later layers.

Max pooling selects the largest value in a local region. Global average pooling calculates the average value of each feature channel across its spatial dimensions.

Softmax Function

In a multi-class classification problem, a model produces scores for different classes. The Softmax function converts these scores into probabilities.

It is defined as:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}} \tag{2}$$

Here, z_i is the score for class i, and K is the total number of classes.

The predicted class is generally the class with the highest output probability.

Residual Connections

A residual connection adds the input of a block to the output of a learned transformation. A simple residual block can be represented as:

$$y = F(x) + x \tag{3}$$

Here, x is the input, $F(x)$ is the transformation learned by the block, and y is the output.

This shortcut allows the block to retain information from its input while learning additional features.

Attention Mechanisms

Attention mechanisms adjust the importance of features. Spatial attention works across image locations, whereas channel attention works across feature channels.

These mechanisms can be useful when some parts of an image are more informative than others. However, their actual benefit depends on the model and the dataset.

Transfer Learning

Transfer learning starts with a model whose weights have already been trained on another dataset. The model can then be adapted to the target task.

One possible approach is to train the new classification layers first and then fine-tune

selected layers of the pretrained network. The exact procedure should match the model implementation being used.

Proposed Methodology

Dataset Used

The project uses the WikiArt-10K dataset. It contains artworks associated with different artistic styles and provides image data and style labels.

The style labels are important because the model needs to know the correct category for each image during training and evaluation.

Dataset Preparation

Before processing the images, the preparation scripts check the dataset files and labels. They also create a mapping between the original style IDs and the numerical labels used by the model.

This mapping helps maintain a consistent relationship between the label numbers and their corresponding artistic styles.

The dataset is divided into three subsets:

- Training set: 70%
- Validation set: 15%
- Test set: 15%

Image Preprocessing

Images can have different dimensions and colour formats. To make them suitable for the model input pipeline, the image preparation script resizes valid images to 224 by 224 pixels.

Each image is converted to RGB format and saved as a JPEG file. The images are organised into folders based on their dataset split and artistic style. The original dataset files remain unchanged.

Proposed Attention-Guided Approach

The proposed approach is to combine CNN feature extraction, residual learning, attention mechanisms, and transfer learning.

The idea is to use a pretrained network to learn image features, refine these features using attention, and use the resulting representation for classification.

The proposed Art-Attention Residual (AAR) block is described as a combination of spatial attention, channel attention, and residual learning. Its actual implementation should be checked in the model source code before claiming that every component was used in the final trained model.

Spatial Attention

Spatial attention gives different weights to locations within a feature map. This may help the model focus on regions containing useful patterns or textures.

Channel Attention

Channel attention adjusts the importance of feature channels. This may help the model make better use of features that are useful for distinguishing artistic styles.

Residual Integration

Residual integration combines learned features with information from the original input. This can help preserve useful information while the network learns additional features.

Transfer Learning Strategy

The original report discusses a two-stage transfer learning strategy.

In the first stage, a pretrained network can be used as a feature extractor while newly added classification layers are trained. In the second stage, selected layers can be unfrozen

and fine-tuned. Fine-tuning usually uses a smaller learning rate to adjust existing weights gradually. The exact layers,learning rates, and training settings must be confirmed from the actual training script.

Experimental Setup

Software and Libraries

The supplied Python files use Python and several libraries for dataset preparation and model evaluation.

The main libraries include:

- **TensorFlow and Keras:** Used to load the saved model and generate predictions.
- **NumPy:** Used to handle numerical arrays and prediction labels.
- **pandas:** Used to read metadata and save evaluation results.
- **scikit-learn:** Used to calculate classification metrics.
- **Pillow:** Used to open, convert, resize, and save images.
- **PyArrow:** Used to read Parquet dataset files.

Evaluation Procedure

The images are loaded, resized to 224 by 224 pixels, and grouped into batches of eight. The model predicts the class probabilities for these images.

The predicted class is compared with the actual label. The script then calculates performance metrics and saves the results in CSV and JSON files.

This evaluation script does not train the model again. It only evaluates the saved model.

Accuracy

Accuracy measures the proportion of predictions that match the correct labels.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{4}$$

A higher accuracy means that a larger proportion of the evaluated images were classified correctly.

However, accuracy alone may not show how well the model performs on every artistic style.

Precision

Precision measures how many predictions for a particular class are correct.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

Here, TP represents true positives and FP represents false positives.

Recall

Recall measures how many actual examples of a class are correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

Here, FN represents false negatives.

F1-Score

The F1-score combines precision and recall into one measure.

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

The supplied evaluation script calculates both macro F1-score and weighted F1-score. Macro F1 gives equal importance to each class, whereas weighted F1 accounts for the number of examples in each class.

Confusion Matrix

A confusion matrix compares the actual class labels with the predicted labels.

It shows how many images from each actual class were assigned to each predicted class. The diagonal entries represent correct predictions, while the other entries show misclassifications.

This can help identify artistic styles that the model finds difficult to distinguish.

Results and Discussion

Evaluation Results

The final evaluation results should be taken from the output of the supplied evaluation script. The script calculates test accuracy, macro F1-score, and weighted F1-score.

It also saves a classification report, a confusion matrix, and a summary of the evaluation results.

The actual values should be inserted after running the script successfully with the saved model and the correct test dataset.

Table 1: Summary of model evaluation

Metric	Obtained value
Test accuracy	To be filled from evaluation output
Macro F1-score	To be filled from evaluation output
Weighted F1-score	To be filled from evaluation output

Classification Report

The classification report provides precision, recall,F1-score, and support for each artistic style.

These values help us understand whether the model performs similarly across different classes. For example, a model may classify some styles accurately while struggling with styles that share similar visual features.

The class-wise report saved by the evaluation script can be used to discuss these differences.

Confusion Matrix Analysis

The confusion matrix provides another way to understand the model's predictions. If two styles are frequently confused with one another, it may indicate that their visual features are difficult for the model to separate.

The matrix should be examined after running the evaluation script. Any discussion of specific styles should be based on the actual matrix rather than an assumption about which styles are difficult.

Limitations and Future Work

Limitations

Art-style classification has several challenges.Different styles can share similar colours, textures, and visual patterns, making classification difficult.

The quality and distribution of the dataset can also affect model performance. If some styles have fewer examples than others, the model may perform differently across classes.

Another limitation is that the supplied evaluation script depends on an existing saved model. Its results depend on the model that was trained and the preprocessing used during training. The evaluation process alone cannot establish how the model was trained or whether every proposed architecture component was implemented.

Future Work

There are several ways in which this project could be extended.

First, the model could be tested on a larger and more varied collection of artworks. This would help assess how well it performs on images with different visual characteristics.

Second, additional baseline models could be evaluated using the same dataset splits. This would make it easier to compare performance under consistent conditions.

A simple interface could be developed to allow users to upload an artwork and view its predicted style,provided the model and its input pipeline support this functionality.

Conclusion

This project explores the use of deep learning for digital art-style classification using the WikiArt-10K dataset. The work includes inspecting the dataset, preparing the images, organising the style labels, and evaluating a saved classification model.

The report also discusses CNNs, residual connections, attention mechanisms, and transfer learning as techniques that may be useful for learning features from artwork images.

The evaluation process uses accuracy, precision, recall,F1-score, and a confusion matrix to study model performance. These measures provide a better understanding of how the model classifies different artistic styles.

Reference

The main reference for this project is the research paper *Improved Deep Convolutional Neural Network for Digital Art Image Classification and Identification* by Huidong Zhang [0].

The paper explores how deep learning can be used to classify digital artworks into different artistic styles. It combines attention mechanisms, residual connections, and transfer learning to improve feature extraction and classification performance. These concepts provide the foundation for understanding the approach studied in our project.

1

H. Zhang, "Improved Deep Convolutional Neural Network for Digital Art Image Classification and Identification," *International Journal of Computational Intelligence Systems*, vol. 18, article 249, 2025. doi: 10.1007/s44196-025-00996-0.